

MSc

Course title: SM6 Bayes Methods
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1. A book club with n members wants to decide what book to read next. They have a shortlist of B books with labels $\mathcal{B} = \{1, \dots, B\}$. Let $\mathcal{P}_{\mathcal{B}}$ be the set of all permutations of the labels in $\mathcal{B}$. For $i = 1, \dots, n$ the i 'th reader gives a ranked list of the books $y_i = (y_{i,1}, \dots, y_{i,B})$, $y_i \in \mathcal{P}_{\mathcal{B}}$, ranking them from most to least interesting. The data are $y = (y_1, \dots, y_n)$.

In a Plackett-Luce model each book $b = 1, \dots, B$ has interest measure $\theta_b > 0$. Let $\theta = (\theta_1, \dots, \theta_B)$, $\theta \in R^B$. Let $Y_i \in \mathcal{P}_{\mathcal{B}}$ denote the random ranking from the i 'th reader. In the Plackett-Luce model, given $Y_{i,1} = y_{i,1}, \dots, Y_{i,a-1} = y_{i,a-1}$, the a 'th entry (ie, the next entry) is decided by choosing book b with probability proportional to θ_b from the books $\mathcal{B} \setminus \{y_{i,1}, \dots, y_{i,a-1}\}$ remaining. The $Y_1, \dots, Y_n$ are jointly independent given θ .

(a) [10 marks]

- (i) Show that the likelihood $L(\theta; y)$ is

$$L(\theta; y) = \prod_{i=1}^n \prod_{a=1}^B \frac{\theta_{y_{i,a}}}{\sum_{b=a}^B \theta_{y_{i,b}}}.$$

- (ii) The prior is $\pi_{\mathcal{B}}(\theta) = \prod_{b=1}^B \pi(\theta_b)$ with $\pi(\theta_b) = \Gamma(\theta_b; \alpha', 1)$ with $\alpha' > 0$ given. Write down the posterior density $\pi(\theta|y)$ and give an MCMC algorithm targeting $\pi(\theta|y)$.
- (iii) Explain why the scale β' in the prior $\Gamma(\alpha', \beta')$ for θ_b , $b \in \mathcal{B}$ may be set equal one. Suppose odds of 1000 : 1 for ranking one book above another represent extreme preference and are a priori unlikely for books on the shortlist. Explain how a fixed numerical value of α' might be chosen, noting any assumptions.

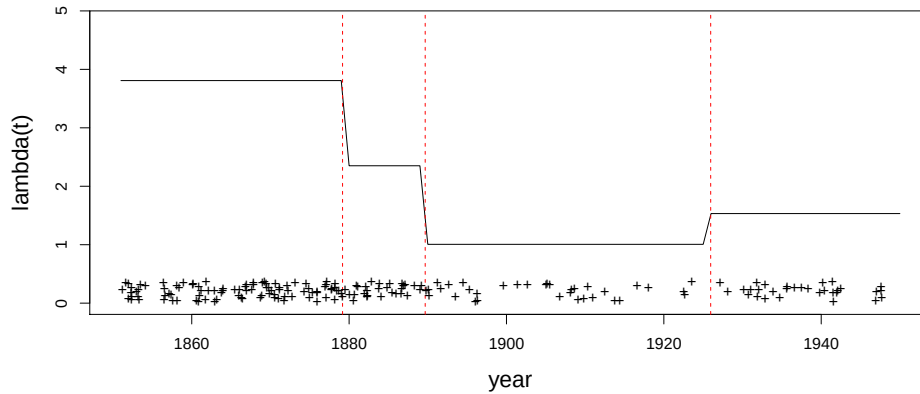
- (b) [10 marks] Suppose B is large so each reader $i = 1, \dots, n$ only reports the first N entries $x_i = (x_{i,1}, \dots, x_{i,N})$ in their ranking, with $N \ll B$. Here $x_{i,j} = y_{i,j}$ for $i = 1, \dots, n$ and $j = 1, \dots, N$. The data are $x = (x_1, \dots, x_n)$.

- (i) Show that the likelihood $L(\theta; x)$ for the new data is

$$L(\theta; x) = \prod_{i=1}^n \prod_{a=1}^N \frac{\theta_{x_{i,a}}}{\sum_{b=a}^N \theta_{x_{i,b}} + \sum_{d \in \mathcal{B} \setminus x_i} \theta_d}.$$

- (ii) Let $\mathcal{C} = \bigcup_{i=1}^n x_i$ give the books appearing in at least one ranking and $\mathcal{D} = \mathcal{B} \setminus \mathcal{C}$ be the books appearing in none. Let $\theta_{\mathcal{C}} = (\theta_b)_{b \in \mathcal{C}}$ and $V = \sum_{d \in \mathcal{D}} \theta_d$. Write down the prior distribution of V and the likelihood $L(\theta_{\mathcal{C}}, V; x)$, and give the posterior $\pi(\theta_{\mathcal{C}}, V|x)$ as a function of $\theta_{\mathcal{C}}$ and V .
- (iii) Give an MCMC algorithm targeting $\pi(\theta_{\mathcal{C}}, V|x)$. State briefly why it may be more efficient, for estimation of $\theta_{\mathcal{C}}$ in the case $|\mathcal{C}| \ll B$, than MCMC targeting $\pi(\theta|x)$.

2. During the 19th and 20th century there were many mining disasters. Let $L = 1850$ and $U = 1950$ and for $i = 1, 2, \dots, n$, let $y_i \in (L, U)$ be the date of the i 'th event. Let $y = (y_1, \dots, y_n)$.



Model the event times y as the arrival times of a Poisson process of piecewise constant rate $\lambda(t)$ per year. Let $\theta_0 = L$ and $\theta_m = U$ and for $i = 1, \dots, m-1$ let $\theta_i \in (L, U)$ be the sorted change point times at which $\lambda(t)$ jumps up or down. For $i = 1, \dots, m$ let $\lambda_i \geq 0$ give the disaster rate over the interval $(\theta_{i-1}, \theta_i]$. The rate function $\lambda(t) = \lambda(t; \theta, \lambda)$ for y is

$$\lambda(t) = \sum_{i=1}^m \lambda_i \mathbb{I}_{\theta_{i-1} < t \leq \theta_i} \quad L < t < U.$$

The data and a realisation of $\lambda(t)$ with $m = 4$ are shown in the Figure, with event dates y (+ signs), change points (θ vertical lines) and $\lambda(t)$ itself (piecewise constant function).

Let $\theta = (\theta_1, \dots, \theta_{m-1})$ and $\lambda = (\lambda_1, \dots, \lambda_m)$. Model the change-point times θ as the arrival times of a Poisson process of unknown rate ρ per year. The number of intervals m is unknown. Prior densities $\pi_R(\rho)$, $\rho \in [0, \infty)$ and $\pi_\Lambda(\lambda|m) = \prod_{i=1}^m \pi_\Lambda(\lambda_i)$, $\lambda \in [0, \infty)^m$ are given.

(a) [8 marks]

- Give a rejection-ABC algorithm sampling $\pi_{ABC}(m, \theta, \lambda, \rho|y)$, the ABC approximation to the posterior $\pi(m, \theta, \lambda, \rho|y)$. Your answer should include summary-statistics and a distance measure. Assume π_R and π_Λ are easily simulated.
- It is of interest to estimate the unknown true rate $\lambda(t)$ as a function of $t \in (L, U)$. State how you would use the ABC-sample output to estimate this function.

(b) [6 marks]

- Write down the prior $\pi(\theta, \lambda, m, \rho)$ in as much detail as you can. Specify its parameter space, $(\theta, \lambda, m, \rho) \in \Omega$ say.
- Write down the posterior $\pi(\lambda, \theta, m, \rho|y)$ in terms of the available model elements.

(c) [6 marks] In a reversible jump MCMC algorithm targeting $\pi(\lambda, \theta, m, \rho|y)$, birth and death updates are chosen with probabilities $p_{m,m+1}$ and $p_{m,m-1}$ respectively. A birth proposal $(\lambda, \theta, m, \rho) \rightarrow (\lambda', \theta', m', \rho)$ with $m' = m+1$ is generated as follows: choose an interval $i \sim U\{1, \dots, m\}$ uniformly; simulate a split point $\theta^* \sim U(\theta_{i-1}, \theta_i)$; simulate two new values $\lambda_{i,1}, \lambda_{i,2} \sim \text{Exp}(1)$ independently. In the candidate state

$$\lambda' = (\lambda_1, \dots, \lambda_{i-1}, \lambda_{i,1}, \lambda_{i,2}, \lambda_{i+1}, \dots, \lambda_m)$$

and

$$\theta' = (\theta_1, \dots, \theta_{i-1}, \theta^*, \theta_i, \dots, \theta_{m-1}).$$

Give a matching death proposal $(\lambda', \theta', m', \rho) \rightarrow (\lambda, \theta, m, \rho)$ and the acceptance probability for the birth proposal. No simplification of expressions is required.